AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING

REPORT
OPEN ENDED LAB

Course no. EEE-2212

Course Title: Measurement and Instrumentation Laboratory.

Name of the Project:

Determination of Applied Force Using Force Sensitive Resistor.

Date of Submission: 26.09.2024

Submitted by

Section: B (B-1)

Group: 04

Student ID: 20220105080, 20220105081, 20220105082,20220105083, 20220105084

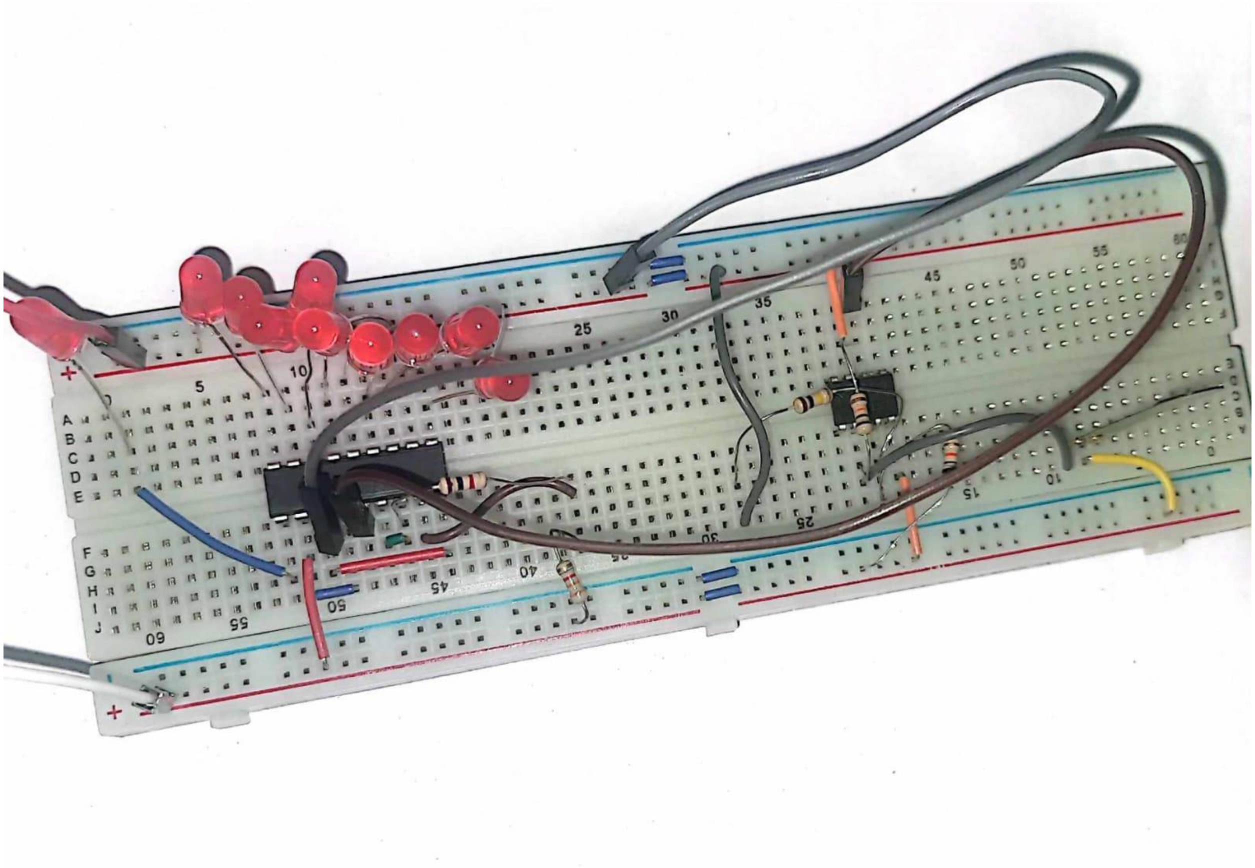
Semester: Fall-2023

Name of The Project: Determination of Applied Force Using Force Sensitive Resistor.

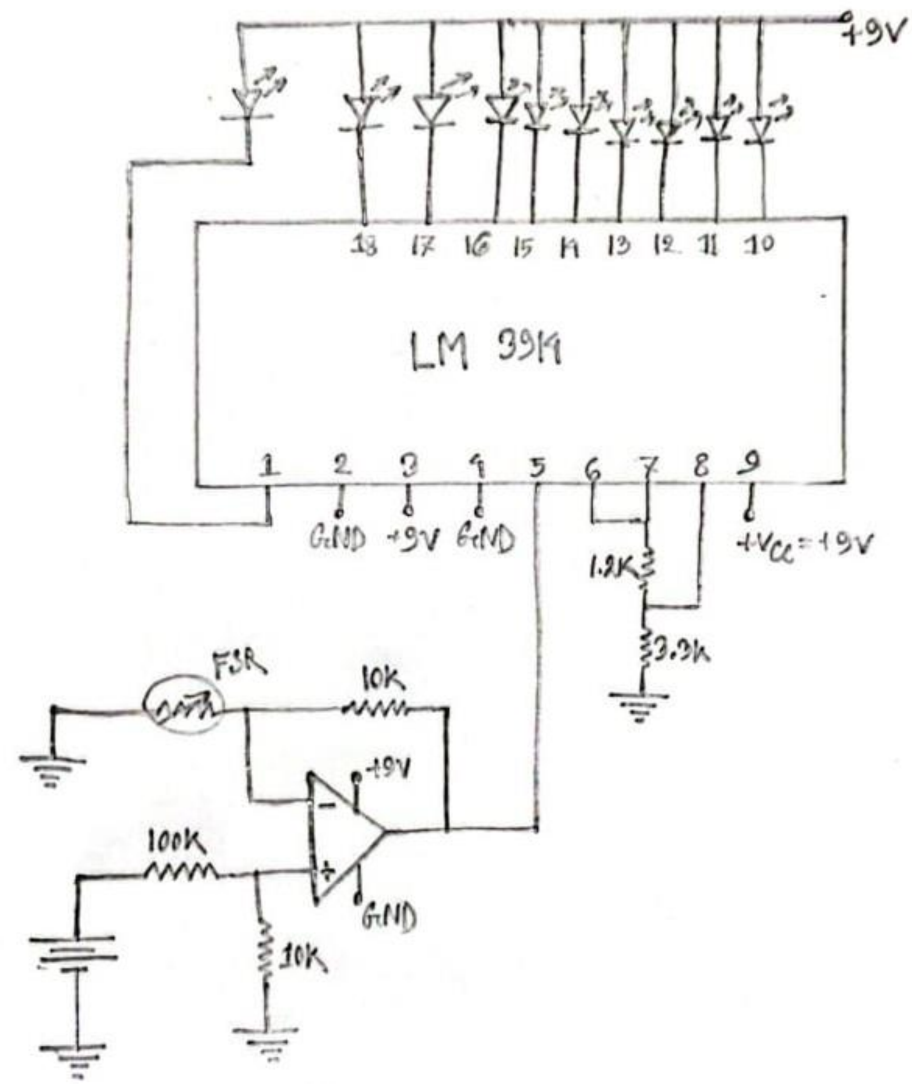
List of Equipment and Price Breakdown:

Name of Components	Quantity	Price
IC- LM3914	1 piece	45
IC-LM741	1 piece	30
Sensor- Force Sensitive Resistor (FSR)	1 piece	525
Breadboard	1 unit	115
Resistors	10kΩ- 2 pieces, 100kΩ- 1 piece, 1.2kΩ- 1 piece, 3.3kΩ- 1 piece	5
LED	10 pieces	16
Connecting Wires	As required	
DC Power Supply	1 unit	

Picture of the Circuit:



Circuit Diagram:



A Section About The Sensor:

Sensor used: Force Sensitive Resistor (SEN-0057)

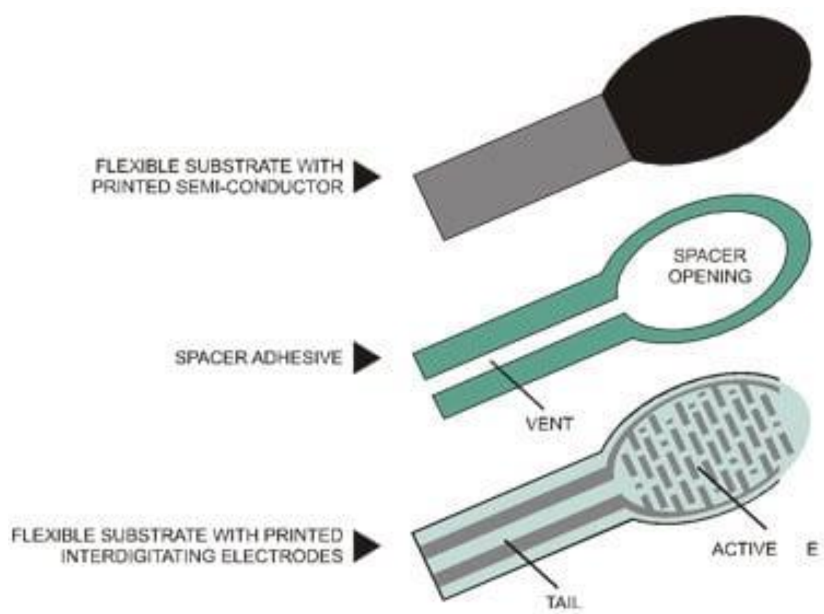
Overall Length: 1.75 in

Overall width: 0.28in

Sensing Diameter: 0.16 in

Force Sensitivity Range: 0.1N to 10N

Internal Construction of The Sensor:



FSR's internal construction consists of two layers separated by a spacer.

Conductive layer (top layer):

- Coated with conductive material i.e. silver, copper etc.
- The conductive pattern plays a crucial role in determining the sensitivity and response of the FSR.

Resistive layer (Bottom Layer);

- Coated with semi-conductor material.

- When force is applied the carbon particles in this layer are pressed together and allow more current to pass through and decrease the overall resistance.

Spacer:

- This layer ensures that there is no contact between conductive and resistive layer in absence of force.
- Contains small air gaps to allow the layers to be in contact when subjected to force.

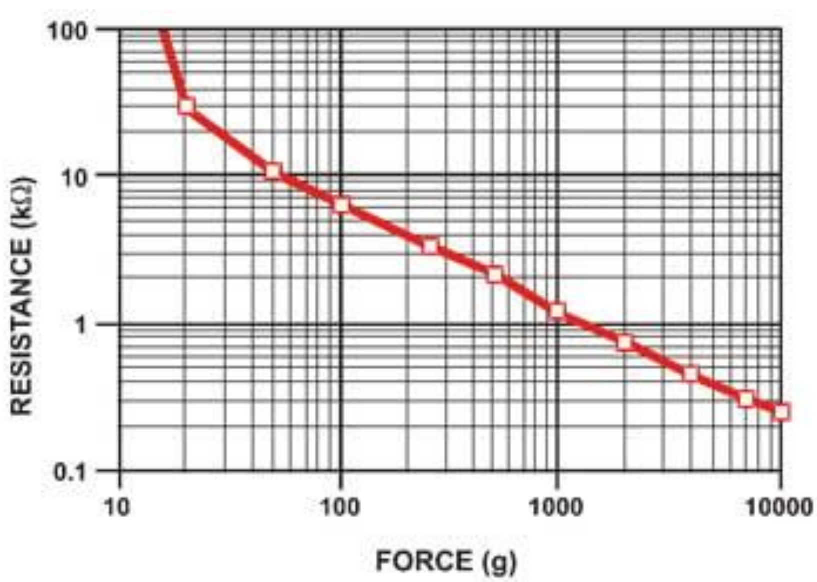
Working Principle of The Entire Circuit:

SENSOR STAGE

When no force is applied, the conductive and resistive layer are not in touch and separated by spacer. So there is no significant pathway for current to be passed through the resistive material and the resistance is very high. (Approximately 100MΩ as measured).

When force is applied the conductive and resistive layer get in touch through the spacer openings or air gaps. As a result an improved pathway is created that allows current to pass through the resistive material and the resistance decreases.

More applied force results in more resistance but the relation is non-linear. The resistance increases in a logarithmic way with the increase of force.



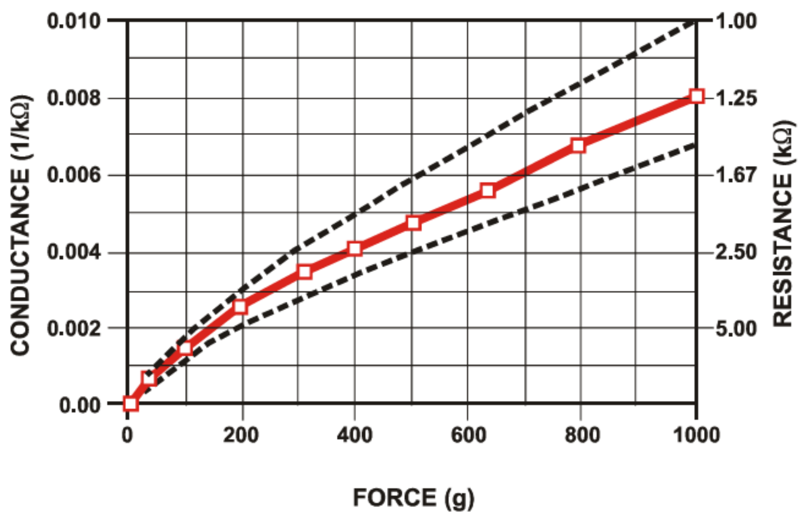


Figure 4:
Conductance vs. Force (0-1Kg) Low Force Range

SIGNAL CONDITIONING STAGE

IC used: µA741; biased at $+V_{cc}$=+9V and $-V_{cc}$=0V

For this stage differential amplifier configuration is utilized.

The voltage at inverting input pin depends on FSR. With applied force, the resistance of FSR changes and the voltage at pin 2 varies too.

Voltage at pin 3 is fixed and is determined by the voltage divider formed by 100kΩ and 10kΩ resistors.

The differential amplifier amplifies the difference between the voltages of inverting and non-inverting pins.
When force is applied to FSR , the resistance of this sensor decreases. The lower this resistance gets, the more voltage is achieved at the output pin.
The output pin is connected to the signal input pin of IC LM3914

OUTPUT STAGE

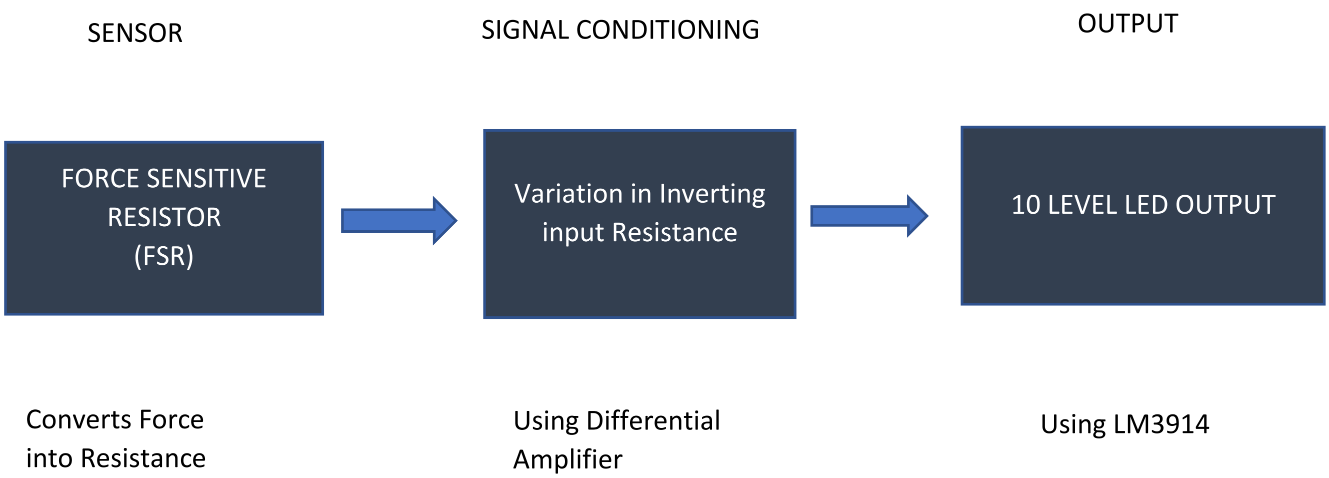
IC: LM3914 in bar mode.

To display the ranges of force applied, LED bar graph is used. At no applied force, the LEDs will remain OFF. But with the increase in applied force, the LEDs will be turned ON according to how much pressure is applied. At full pressure, all the LEDs will be turned on. Thus applied force can be approximated by observing the LED display.

Questions:

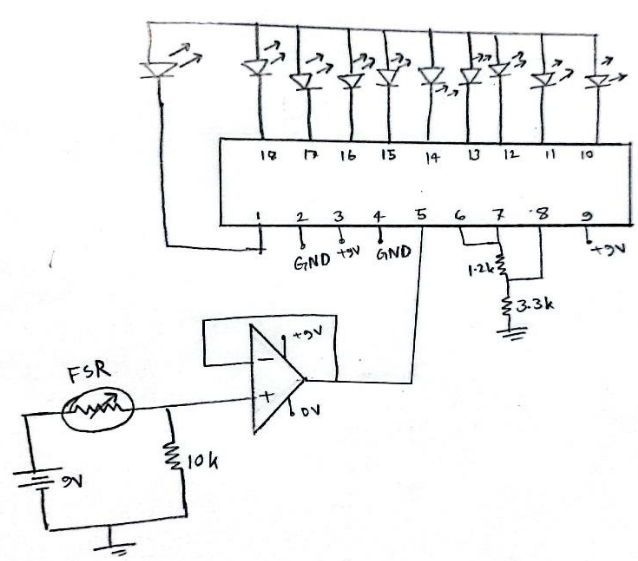
1. Consider the project comprise of three sub-sections (Sensor, Signal conditioning and Output) then create the block diagram of the project.

Answer:



2. Draw circuit diagram of an alternate circuit that use the same sensor.

Answer: here is a buffer circuit using Force Sensitive Resistor.



Reference:

- https://learn.adafruit.com/force-sensitive-resistor-fsr/overview
- FSR Datasheet